

Home automation may also allow vital home functions to be controlled remotely from anywhere in the world, using any device with a browser that has an Internet connection.

A smart home is a wonderful thing. It learns your habits, watches out for intruders and even sends you alerts when things don't go according to plan. And it communicates with you remotely.

Trustworthy and powerful communication protocols like Z-wave or Zigbee still work with open source systems. The hubs that work with open source smart home software are competitively priced, and are perhaps even cheaper in the long run because one can optimise home systems more easily with them.

A list of nine open source solutions available in the market, with their main features, is given below.

Home Assistant

This open source home automation platform has been developed with software written in Python. It works asynchronously and has very low requirements in terms of specifications. It can control a home's lights and window blinds using a programmable logic controller connected to a home assistant using the MQTT protocol, and it can receive information from several sensors using an RF gateway. It can also be connected to IP cameras to retrieve motion sensing events and send notifications through various services.

Open Hab

This home automation platform aims for the integration and interaction between different home technologies and devices.

Main features:

- Vendor- and technology-agnostic
- Developed in Java
- Runs on any device capable of running JVM (Java Virtual Machine)
- Integrates different home automation technologies into one
- Comes with different Web based user interfaces (UI)
- Provides APIs for being integrated with other systems

MyController

This home automation server allows the monitoring and control of the home from anywhere.

Main features:

- Very lightweight server
- Supports Linux, Windows and Raspberry Pi
- Supports multiple gateways with many protocols – serial, Ethernet and MQTT
- Time based operations

- Gives notifications, alarms and graphical reports
- Comes with interactive dashboards

Domoticz

This is a lightweight home automation system that allows the monitoring and configuration of lights, switches, sensors and other devices. It also allows you to send notifications and alerts to any mobile device.

Main features:

- Compatible with all browsers
- Works on several operating systems with a scalable HTML 5 Web front-end that adapts to desktop and mobile devices
- Runs on Raspberry Pi, the Cubie board, UNIX, Apple and Windows

Fhem

This is a Perl server for home automation, released under a GPL licence. It automates tasks like switching on lamps, window shutters, heating and so on. It also logs events like temperature, humidity and power consumption.

Main features:

- Runs as a server
- Controls devices via the Web or the smartphone front-end, and via Telnet or TCP/IP directly
- Supports several protocols
- Auto creates a device's logs when receiving data from a new device
- Logs events to files or the database
- Sends notifications about external programs or scripts when it receives data about certain events
- Timed commands
- Several interfaces: Simple text, JSON, XML, etc

Major DoMo

This is an open source home automation platform designed to be used in a multi-protocol and multi-services environment.

Main features:

- Can be installed on almost any computer running Windows or Linux OS
- Easy installation
- Supports many protocols (MQTT, Zwave, I-wave, I-wore, ModBus, SNMP, Ethernet)
- Offers Web based interfaces for any device
- GPS-tracking
- Scripting with PHP and/or Blockly
- Sets up automation rules and scripts

Pytomatic

This is a lightweight open source automation system written